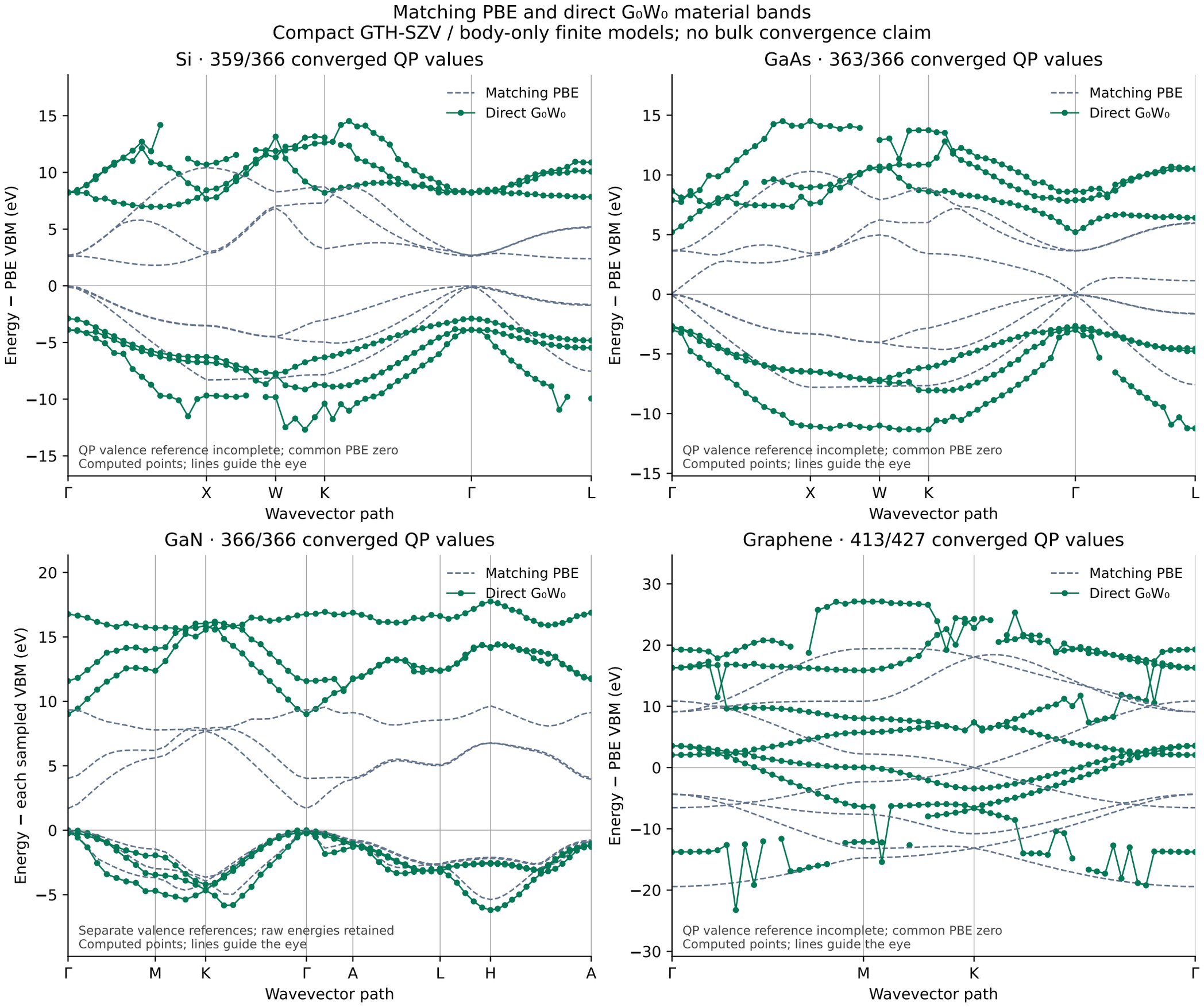

Matching PBE and direct G₀W₀ material bands
Compact GTH-SZV / body-only finite models; no bulk convergence claim
Si · 359/366 converged QP values
GaAs · 363/366 converged QP values
GaN · 366/366 converged QP values
Graphene · 413/427 converged QP values
Matching PBE
Direct G₀W₀
Energy − PBE VBM (eV)
Energy − each sampled VBM (eV)
QP valence reference incomplete; common PBE zero
Separate valence references; raw energies retained
Computed points; lines guide the eye
Γ X W K Γ L
Γ M K Γ A L H A
Γ M K Γ
Wavevector path